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TITLE PAGE

- **Problem Statement ID – 25233**
- **Problem Statement Title - Automated Vulnerability Assessment and Penetration Testing tool for CCTV cameras & DVRs**
- **Theme - Smart Automation**
- **PS Category - Software**
- **Team ID - 56984**
- **Team Name (Registered on portal) - Xyndrix**



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Automated Vulnerability Assessment and Penetration Testing tool for CCTV cameras & DVRs

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PROBLEM STATEMENT ANALYSIS



Large and vulnerable attack surface



Significant Security Risk



Lack of tools for CCTV/DVR systems

WHAT IS NEEDED?

Needed: Automated CCTV security with ML, predictive analysis, and CVE mapping

Outcome: All-in-one interactive tool for device discovery, vulnerability reports, exploits, and fixes

Core focus



VA/PT Tools



ML algorithms



Automation & Scalability

PROPOSED SOLUTION

VAPT Automation

A scalable tool for automated VAPT of CCTVs and DVRs

Global Threat Visualization

This tool provides a visual dashboard showing a worldwide map of discovered devices

Pinpoint Threats with Precision

This tool can target global scanning devices using IP ranges, device models, etc.

ML-Driven Mitigation Guidance

ML-driven recommendation engine that provides steps for mitigating all identified risks

UNIQUENESS

AI-Powered Vulnerability Analysis

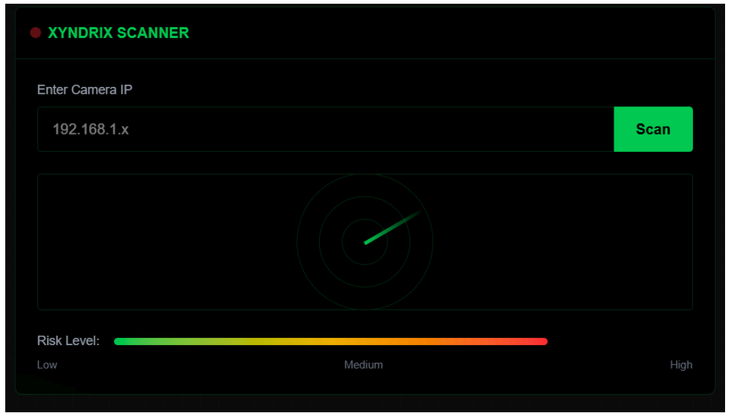
Machine learning detects anomalies, predicts risks, and links them to known CVEs

Smart Video Transcription

Transcribes CCTV video feeds into searchable text, enabling deeper threat intelligence and activity analysis

Automated Global Scanning

Capable of discovering, fingerprinting, and testing thousands of cameras worldwide

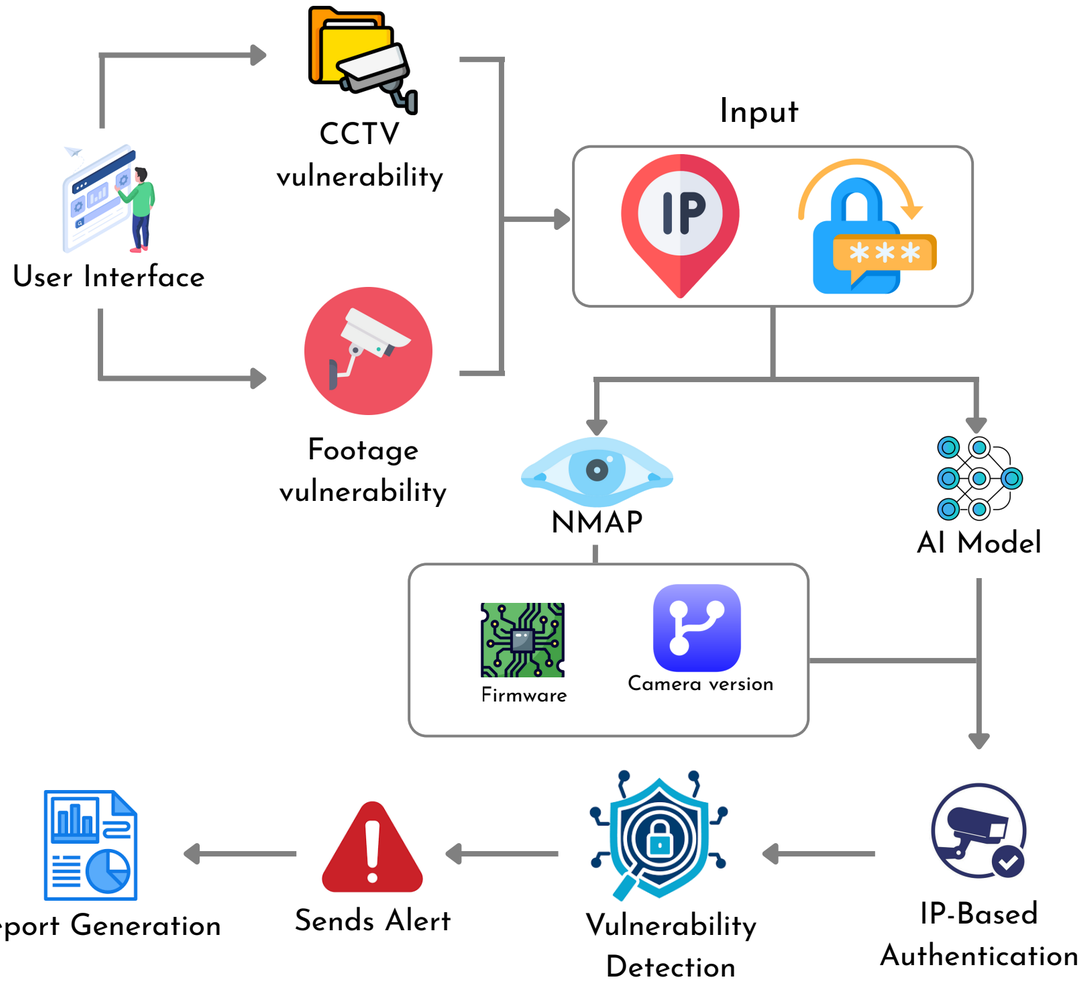


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TECHNICAL APPROACH

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WORKFLOW



FOOTAGE DETECTION



FRONTEND	Fast, clean, responsive dashboard for CCTV vulnerability monitoring	NEXT.JS Tailwind CSS
BACKEND	Secure, high-speed backend for data processing and authentication	FastAPI django
SCANNING TOOLS	Scans CCTV networks to detect and report vulnerabilities	
DATABASE	Stores and manages CCTV scan data efficiently and securely	PostgreSQL

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FEASIBILITY AND VIABILITY

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Technical Feasibility – Existing tools like Nmap, Shodan, and Metasploit can be integrated, and ML can be applied to vulnerability prediction.

Operational Feasibility – Security professionals can easily adopt it with a dashboard and automation features.

Data Feasibility – Public CVE databases, firmware info, and honeypot data can be leveraged to train ML models and enrich vulnerability mapping.

CHALLENGES	STRATEGIES
Diversity of CCTV/DVR brands, models, and firmware → hard to cover all vulnerabilities.	Classify diverse device models by their common architecture for targeted testing.
Lack of large labeled datasets for training ML models.	Leverage honeypots, threat feeds, synthetic data, and vendor collaboration.
Legal and Ethical Issues in Global Scanning	Restrict to authorized scans and adopt government-approved frameworks.

Automated Vulnerability Scanning

Penetration Testing Simulation

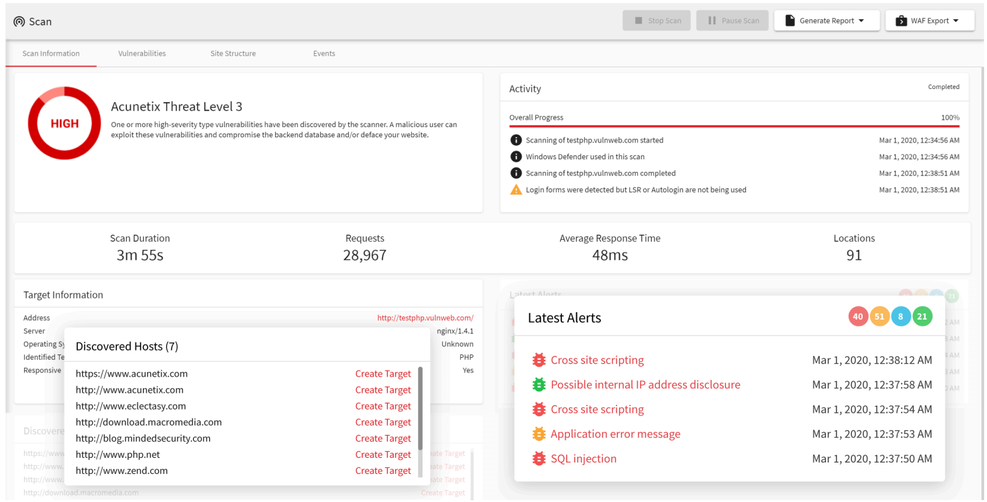
ML-Based Threat Prediction

SCOPE

Global Device Dashboard

CVE Vulnerability Mapping

Mitigation Risk Recommendations



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IMPACT AND BENEFITS

IMPACT

Up to 40% Faster Vulnerability Detection - Automated scanning reduces manual testing time.

30-35% Reduction in Security Risks - Proactive detection of flaws lowers chances of exploitation.

25% Improvement in Incident Response Efficiency - Faster patching enabled by mapped CVEs.

20% Cost Savings in Security Audits - Reduced reliance on repetitive manual assessments.

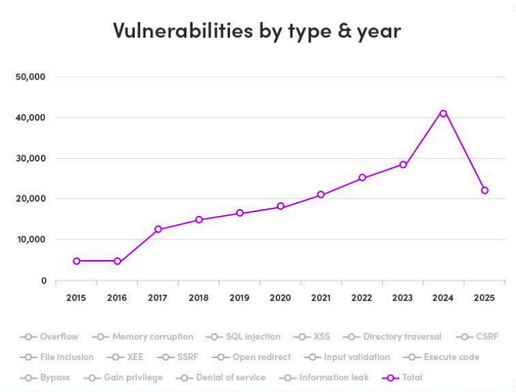
BENEFITS

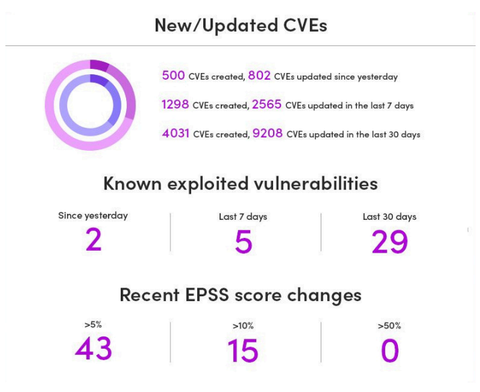
Data-Driven Decision Making
ML insights improve prioritization of vulnerabilities.

User-Friendly Adoption
Security professionals can operate it easily without deep technical expertise.

Future-Proofing
Adaptive ML models evolve with new threats and zero-day vulnerabilities.

Global Device Visibility
Dashboard enables mapping of CCTV/DVR devices worldwide.





Source: Data aggregated from public CVE databases (NIST NVD, MITRE)

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RESEARCH AND REFERENCES

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1. Vulnerability Assessment and Penetration Testing on IP Camera – Case study on scanning and exploiting vulnerabilities in IP cameras. **Link here**
2. Evaluating IP Surveillance Camera Vulnerabilities – Analysis of real-world camera weaknesses (default credentials, poor configs). **Link here**
3. Unmasking Vulnerabilities by a Pioneering Approach to Smart IoT Cameras – 2024 framework for smart surveillance devices. **Link here**
4. Risk Assessment of an Installed CCTV System in an Open Marketplace – Real-world deployment vulnerabilities and risks. **Link here**
5. Deep Reinforcement Learning for Automated Penetration Testing – Using AI/DRL for automating attack path discovery. **Link here**
6. HARMer Framework – Hierarchical attack representation model for cyber-attack automation. **Link here**
7. Autosploit: Automated Exploitability Framework – Automates testing of whether vulnerabilities can actually be exploited. **Link here**
8. VulnBot: Multi-Agent LLM Penetration Testing System – Shows how AI agents can coordinate scanning, exploitation, reporting. **Link here**
9. PentestAgent: LLM Agents for Automated Penetration Testing – Focused on automating reconnaissance, exploitation, and reporting. **Link here**
10. Critical Flaws in Dahua CCTV Cameras – Industry case showing real-world remote takeover vulnerabilities. **Link here**